LEVEL 1: EASY (Linear Equations & 1-2 Step Isolations)

Exercise 1

Isolate the variable x:

$$4x - 9 = 15$$

Answer 1

1. Undo subtraction by adding 9 to both sides:

$$4x = 15 + 9 \implies 4x = 24$$

2. Undo multiplication by dividing by 4:

$$x = \frac{24}{4}$$

x = 6

Exercise 2

Isolate the variable x in terms of y and b:

$$y = 3x + b$$

Answer 2

1. Undo addition by subtracting b from both sides:

$$y - b = 3x$$

2. Undo multiplication by dividing by 3:

$$\frac{y - b}{3} = x$$

x = $\frac{y-b}{3}$

Exercise 3

Convert the exponential equation to logarithmic form to solve for x:

$$3^x = 81$$

Answer 3

1. Apply base-3 logarithm ($\log_3$) to both sides:

$$\log_3(3^x) = \log_3(81)$$

2. Simplify using the inverse property $\log_b(b^k) = k$:

$$x = \log_3(3^4)$$

x = 4

Exercise 4

Isolate the variable x:

$$\frac{x}{6} + 4 = 9$$

Answer 4

1. Undo addition by subtracting 4 from both sides:

$$\frac{x}{6} = 9 - 4 \implies \frac{x}{6} = 5$$

2. Undo division by multiplying both sides by 6:

$$x = 5 \cdot 6$$

x = 30

Exercise 5

Convert the logarithmic equation to exponential form to solve for x:

$$\log_2(x + 3) = 5$$

Answer 5

1. Exponentiate both sides with base 2:

$$2^{\log_2(x+3)} = 2^5 \implies x + 3 = 32$$

2. Subtract 3 from both sides:

$$x = 32 - 3$$

x = 29

LEVEL 2: INTERMEDIATE (Powers, Roots & General Base Exponentials)

Exercise 6

Isolate the positive variable x:

$$2x^2 - 5 = 45$$

Answer 6

1. Undo subtraction by adding 5:

$$2x^2 = 50$$

2. Undo multiplication by dividing by 2:

$$x^2 = 25$$

3. Undo square power with square root:

$$x = \sqrt{25}$$

x = 5

Exercise 7

Solve for x by applying logarithm base 5:

$$3 \cdot 5^{2x} = 75$$

Answer 7

1. Divide both sides by 3:

$$5^{2x} = 25$$

2. Apply logarithm base 5 ($\log_5$) to both sides:

$$\log_5(5^{2x}) = \log_5(25) \implies 2x = 2$$

3. Divide by 2:

x = 1

Exercise 8

Isolate the variable x:

$$\sqrt{3x+1} = 5$$

Answer 8

1. Undo square root by squaring both sides:

$$(\sqrt{3x+1})^2 = 5^2 \implies 3x + 1 = 25$$

2. Subtract 1:

$$3x = 24$$

3. Divide by 3:

x = 8

Exercise 9

Isolate x from the logarithmic equation:

$$\log_4(2x-6)+1=3$$

Answer 9

1. Subtract 1 from both sides:

$$\log_4(2x-6)=2$$

2. Exponentiate both sides with base 4:

$$4^{\log_4(2x-6)}=4^2 \implies 2x-6=16$$

3. Add 6 and divide by 2:

$$2x=22 \implies x=11$$

x = 11

Exercise 10

Solve for x by applying logarithm base 2:

$$7\cdot 2^{3x}=42$$

Answer 10

1. Divide both sides by 7:

$$2^{3x}=6$$

2. Apply logarithm base 2 ($\log_2$) to both sides:

$$\log_2(2^{3x})=\log_2(6) \implies 3x=\log_2(6)$$

3. Divide by 3:

$$\mathbf{x=\frac{\log_2(6)}{3}}$$

LEVEL 3: HARD (Algebraic Fractions, Physics Formulas & Arbitrary Base Equations)

Exercise 11

Isolate x from the kinematic formula:

$$v^2 = u^2 + 2ax$$

Answer 11

1. Undo addition by subtracting u^2 from both sides:

$$v^2 - u^2 = 2ax$$

2. Undo multiplication by dividing by $2a$:

$$\frac{v^2 - u^2}{2a} = x$$

$$\mathbf{x = \frac{v^2 - u^2}{2a}}$$

Exercise 12

Isolate x when it appears on both numerator and denominator:

$$y = \frac{2x + 1}{x - 3}$$

Answer 12

1. Multiply both sides by $(x - 3)$:

$$y(x - 3) = 2x + 1 \implies yx - 3y = 2x + 1$$

2. Collect terms with x on one side and non-x terms on the other:

$$yx - 2x = 3y + 1$$

3. Factor out x and divide:

$$x(y - 2) = 3y + 1$$

$$\mathbf{x = \frac{3y + 1}{y - 2}}$$

Exercise 13

Isolate x from the period of a pendulum formula:

$$T = 2\pi\sqrt{\frac{x}{g}}$$

Answer 13

1. Divide both sides by 2π:

$$\frac{T}{2\pi} = \sqrt{\frac{x}{g}}$$

2. Square both sides to eliminate the radical:

$$\left(\frac{T}{2\pi}\right)^2 = \frac{x}{g} \implies \frac{T^2}{4\pi^2} = \frac{x}{g}$$

3. Multiply by g:

$$\mathbf{x = \frac{gT^2}{4\pi^2}}$$

Exercise 14

Solve for x in the multi-step exponential equation with base 10:

$$4 \cdot 10^{3x-1} = 200$$

Answer 14

1. Divide both sides by 4:

$$10^{3x-1} = 50$$

2. Apply logarithm base 10 (log) to both sides:

$$3x - 1 = \log(50)$$

3. Add 1 and divide by 3:

$$3x = \log(50) + 1$$

$$\mathbf{x = \frac{\log(50)+1}{3}}$$

Exercise 15

Solve for x by applying logarithm base 3:

$$3^{2x+1} = 7^x$$

Answer 15

1. Apply logarithm base 3 ($\log_3$) to both sides:

$$2x + 1 = \log_3(7^x) \implies 2x + 1 = x \log_3(7)$$

2. Group terms containing x on one side:

$$2x - x \log_3(7) = -1$$

3. Factor out x and isolate:

$$x(2 - \log_3(7)) = -1$$

$$\mathbf{x = \frac{-1}{2-\log_3(7)} = \frac{1}{\log_3(7)-2}}$$